

Retail Customer Intelligence and Retention Analytics

Internship Project Report

Data Science Internship at DMV CoreTech

Field	Details
Submitted by	Pothamsetti Venkata Rama Krishna Reddy
Programme	B.Tech CSE - Data Science
Student ID / Roll No.	23CS002715
Institution	Sir Padampat Singhania University, Udaipur
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Abstract

This project converts Online Retail II transaction data into a reproducible retail customer intelligence and retention analytics system. The work covers data acquisition, raw data validation, cleaning, transaction classification, SQL modelling, exploratory data analysis, RFM customer segmentation, cohort-style retention monitoring, clustering, dashboard exports, and Streamlit deployment.

The final validated pipeline confirms 1,007,913 completed-sales lines, GBP 20,476,260.45 in completed sales, 40,077 completed orders, 5,878 identified customers, and a 72.39% repeat customer rate. The customer intelligence layer further identifies RFM segments and K-Means customer clusters for retention and targeting decisions.

Internship Project Context

This project was developed as an individual project during my Data Science Internship at DMV CoreTech from June 2026 to July 2026. It demonstrates the practical application of data analytics, customer intelligence, feature engineering, visualization, SQL analysis, customer segmentation, and dashboard development to solve a retail business analytics problem.

The project is presented as an internship learning and implementation project, not as an official commercial product of DMV CoreTech.

1. Introduction

Retail businesses generate large volumes of transaction data, but raw transactions alone do not directly explain customer behaviour or business performance. This June 2026 - July 2026 internship project builds an analytical layer that converts transaction records into validated metrics, customer segments, retention views, and dashboard-ready insights.

2. Business Problem

A retail business may have detailed transaction history but still lack a reliable customer intelligence layer. Without proper analytics, stakeholders may struggle to identify high-value customers, inactive customers, repeat purchasing behaviour, revenue concentration, return patterns, product demand, and retention opportunities.

3. Project Objectives

1. Acquire and validate the Online Retail II dataset from the official source.
2. Clean transaction data and separate completed sales, returns, anonymous transactions, exclusions, and outlier flags.
3. Create a SQLite star schema for reproducible SQL-based business analysis.
4. Generate sales, customer, product, geography, returns, and time-based analytics.
5. Calculate customer intelligence features and actionable customer segments.
6. Measure retention behaviour and repeat-purchase patterns.
7. Export dashboard-ready outputs and deploy a Streamlit app for interactive viewing.
8. Translate analytical findings into practical retail recommendations.

4. Dataset Overview

The project uses the Online Retail II dataset from the UCI Machine Learning Repository. It contains transaction records for a UK-based online retail business between December 2009 and December 2011.

Metric	Value
Dataset	Online Retail II
Source	UCI Machine Learning Repository
Raw transaction lines	1,067,371
Expected unique transaction lines	1,033,036
Date coverage	1 Dec 2009 to 9 Dec 2011
Customers	5,942
Products	5,304
Countries	43
Invoices	53,628
License	CC BY 4.0

5. Tools and Technologies

- Python, Pandas, and NumPy for data cleaning, feature engineering, and analytics.
- SQLite and SQLAlchemy for star schema modelling and reproducible SQL analysis.
- SciPy and Scikit-learn for statistical analysis and clustering evaluation.
- Matplotlib, Seaborn, Plotly, Power BI, and Streamlit for visualization and dashboard delivery.
- GitHub Actions and Pytest for validation, testing, and project reliability.

6. Methodology

Phase	Description
Data acquisition	Download and load the official Online Retail II workbook.
Raw validation	Check schema, dates, duplicates, missing values, customer IDs, invoices, products, and countries.
Cleaning	Remove exact duplicates, classify sales and returns, preserve exclusions, and flag outliers.
SQL modelling	Build a SQLite star schema with fact and dimension tables for reliable business querying.
EDA	Create visual analysis for sales trends, product demand, geography, returns, and seasonality.
Customer analytics	Calculate customer value features, RFM-style groups, inactivity risk, and retention views.
Statistics and clustering	Apply hypothesis testing and K-Means-style clustering evaluation to compare business-defined segments with data-driven customer groups.
Dashboarding	Export clean tables and build dashboard views for business users.
Deployment	Publish the Streamlit app and maintain the project on GitHub.

7. Data Quality and Cleaning

Data quality was handled as a separate analytical phase because retail transaction data contains duplicates, cancellations, missing customer identifiers, and outlier values. The project avoids blindly deleting suspicious records and instead documents each decision.

Issue / Decision	Verified Number or Reason
Exact duplicate records	34,335 duplicates across the workbook (3.22% of raw lines).
Completed-sale rows	1,041,670 rows before deduplication.
Return/cancellation rows	22,951 rows before deduplication.
Anonymous rows	243,007 rows (22.77%); retained for sales analysis but excluded from customer-level RFM/cohort analysis.
Missing descriptions	4,382 rows (0.41%).
Final month	December 2011 is partial because the source data ends on 2011-12-09.

8. Results and Visualizations

The following visuals summarize the most important verified business, customer intelligence, retention, and revenue-concentration results. They are taken from the project's generated chart outputs and support the SQL, RFM, cohort, and dashboard analysis layers.

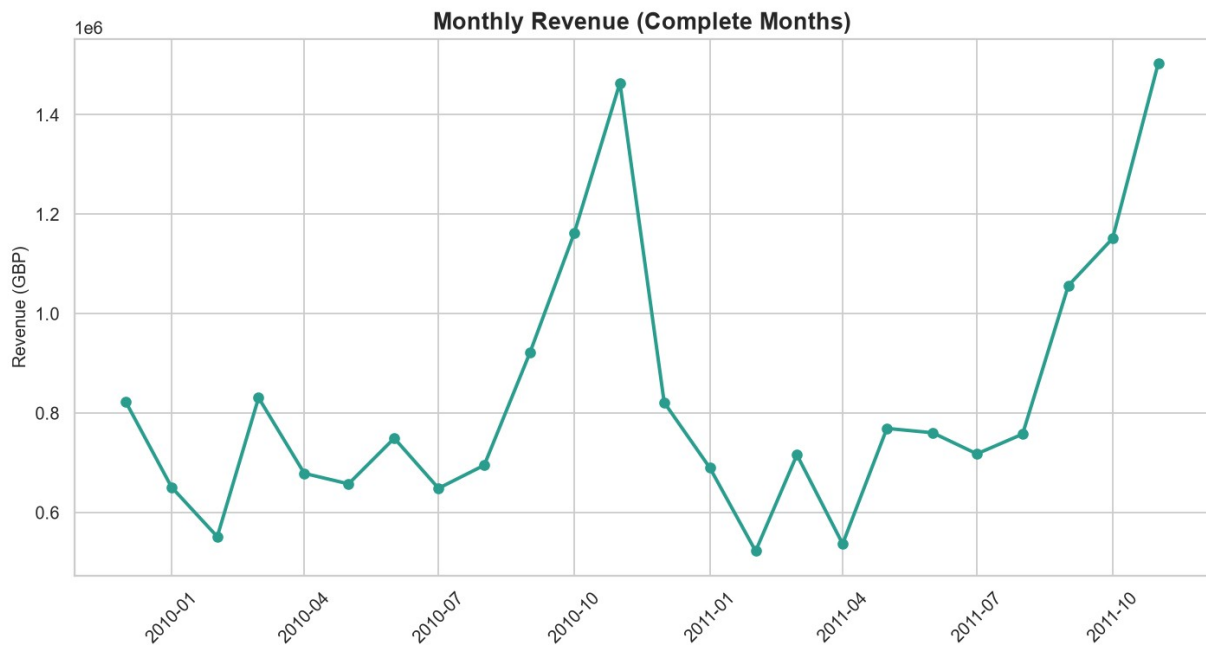


Figure 1: Monthly revenue trend across complete reporting months.

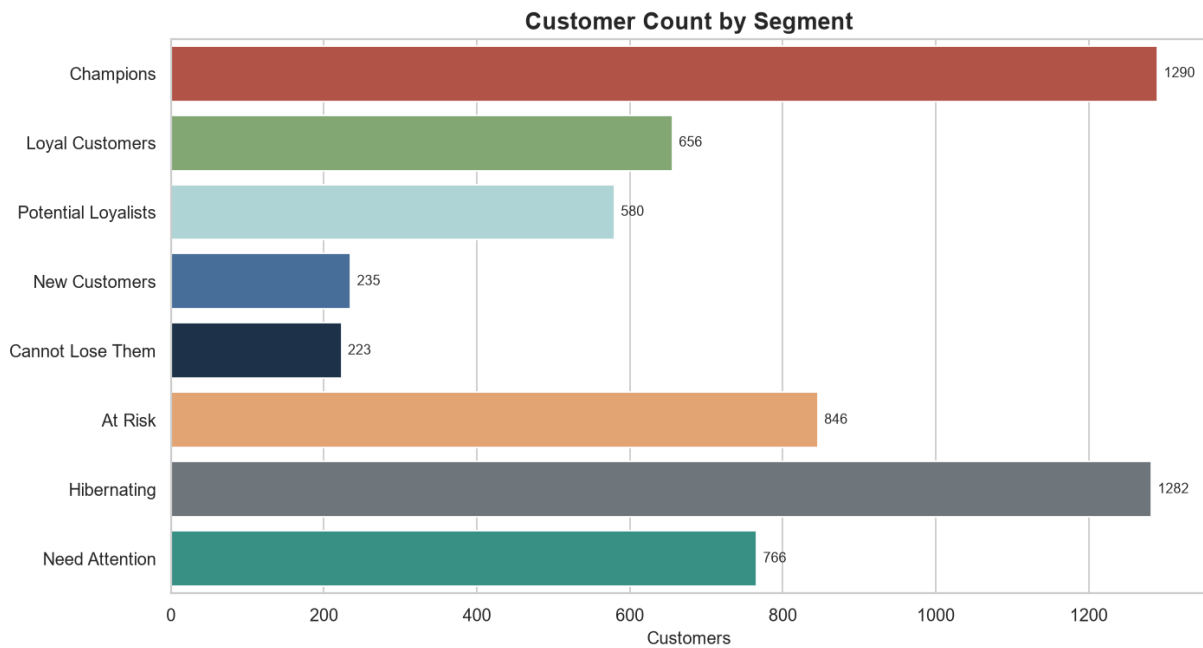


Figure 2: RFM customer segment distribution.

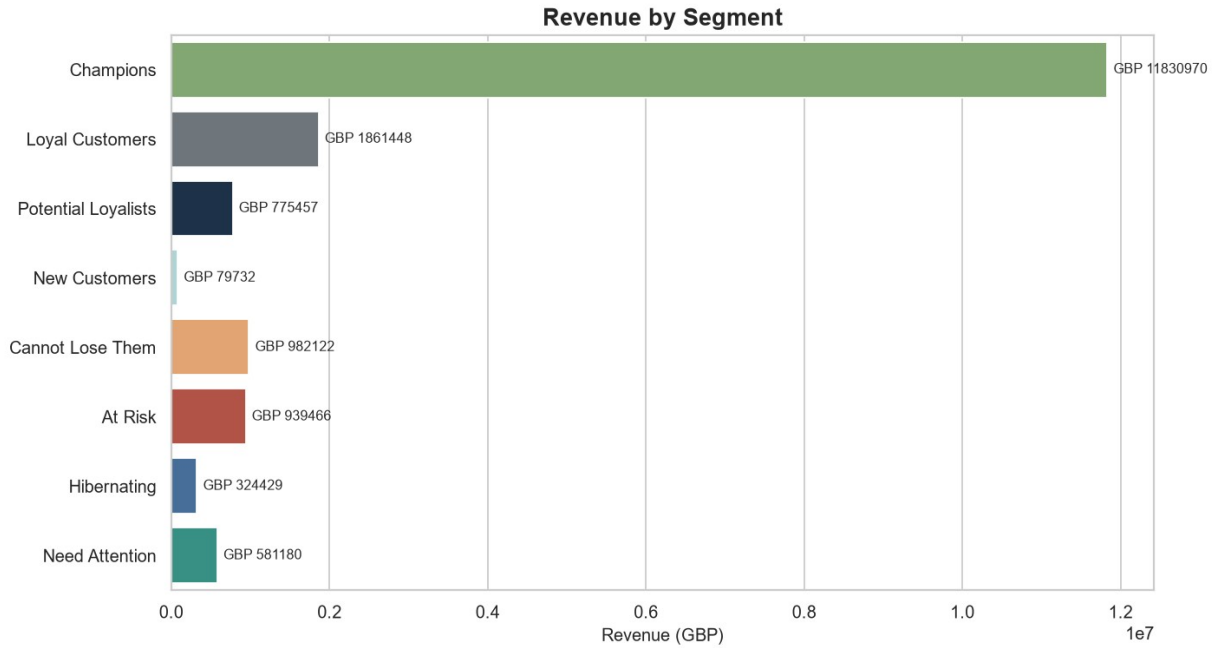


Figure 3: Revenue contribution by RFM customer segment.

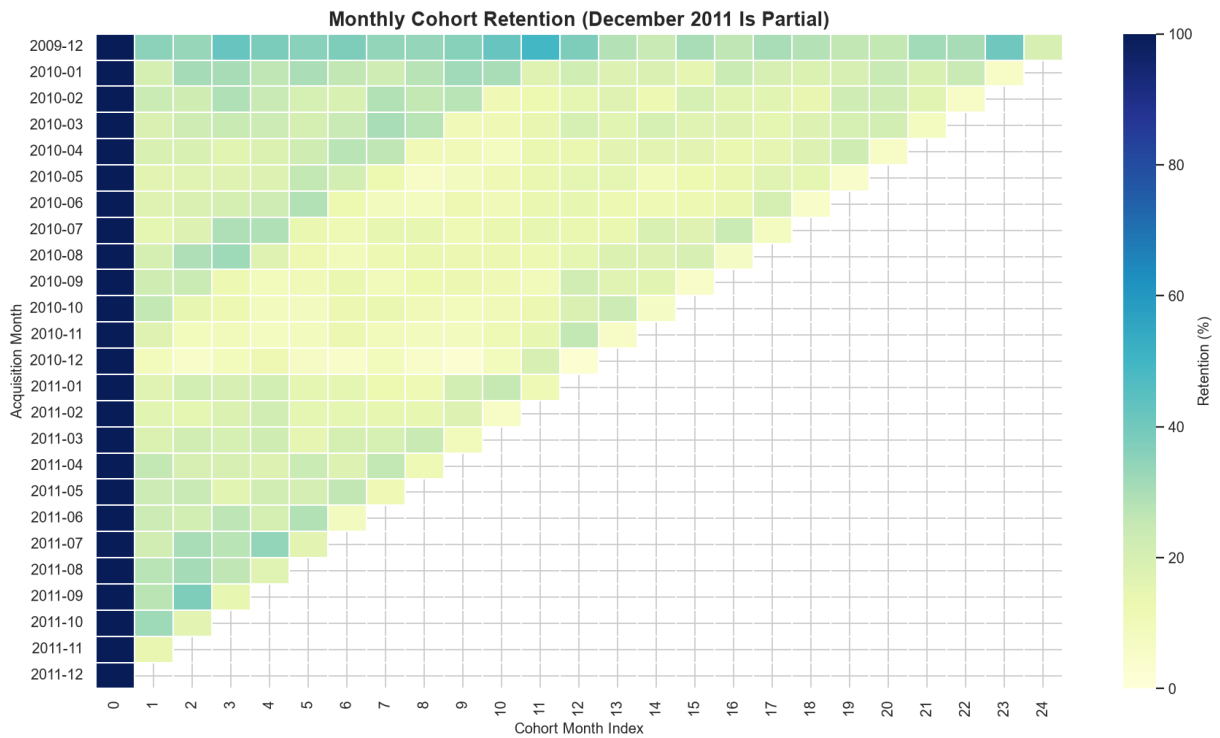


Figure 4: Cohort retention heatmap showing month-wise customer retention.

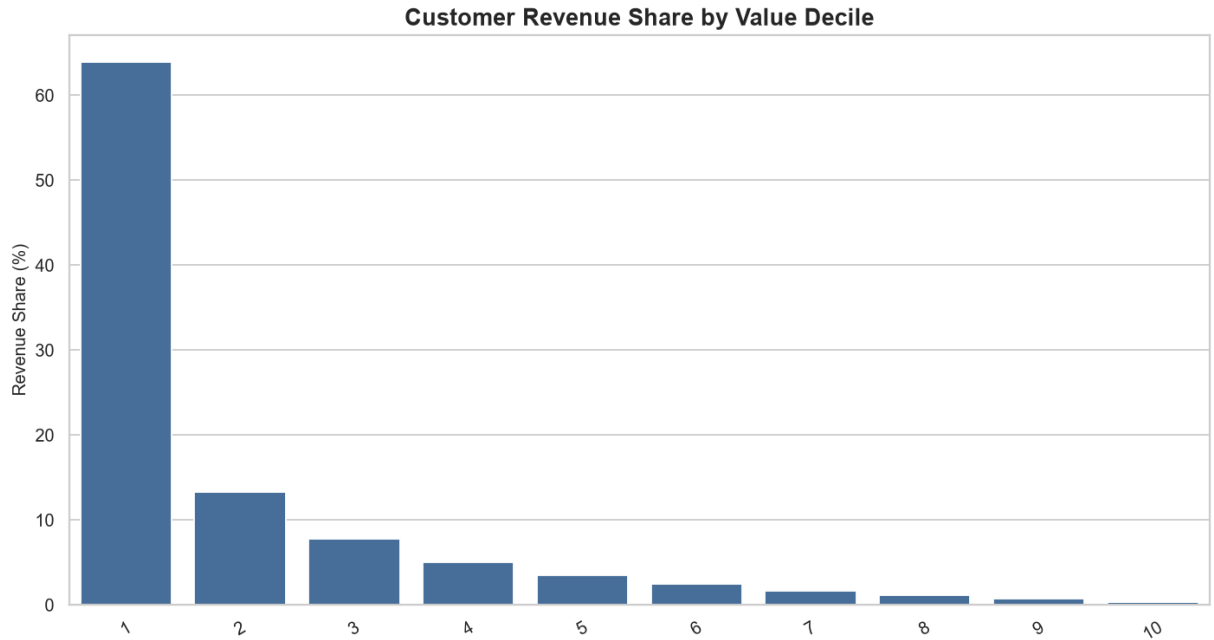
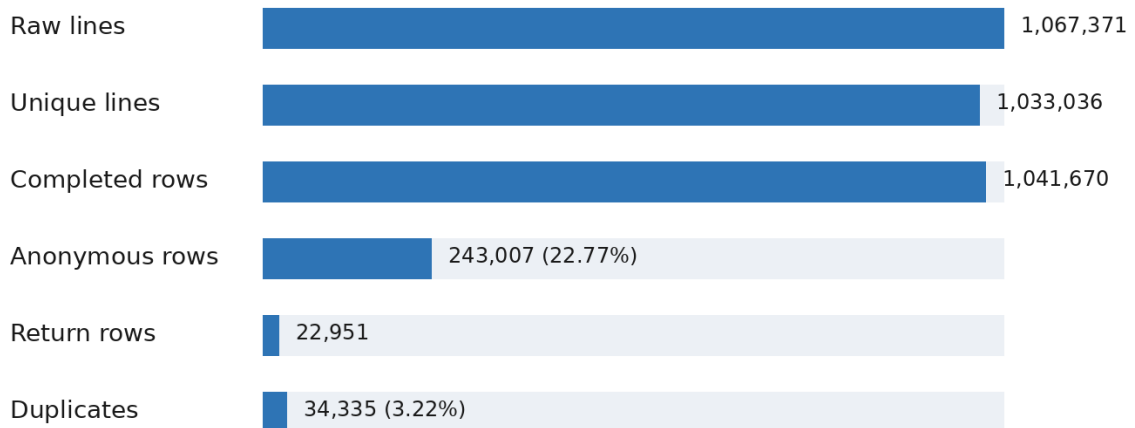


Figure 5: Customer revenue deciles showing value concentration.

Data Volume and Quality Profile



Source: project raw data-quality review.

Figure 6: Data volume and quality profile.

9. Analytical Outputs

The SQL-backed CSV outputs cover decision areas such as monthly revenue growth, country-wise sales, product performance, top customers, repeat-customer rate, customer revenue deciles, return summaries, and data-quality reconciliation. The chart outputs summarize monthly revenue, completed orders, top countries, top products, customer revenue distribution, revenue deciles, identified versus anonymous revenue, outlier revenue impact, RFM segments, and customer clusters.

The customer analytics layer also supports RFM-style segmentation, cohort-style retention monitoring, and clustering comparison. These outputs are used to separate aggregate sales reporting from customer-level retention and segmentation analysis, especially because anonymous transactions cannot be used for customer-level metrics.

Output Area	Result
SQL business outputs	13 CSV outputs for revenue, customers, products, geography, returns, and quality checks.
Exploratory data analysis	14 PNG charts and 2 Plotly HTML charts for trends, distributions, and dashboard review.
Customer analytics	RFM segment table, segment action plan, cohort-style retention outputs, and customer intelligence summary.
K-Means clustering	K=4 operational customer clustering with silhouette 0.366 and five seed mean ARI 0.997.
Dashboard exports	Power BI file, PDF, and screenshots for stakeholder presentation.
Streamlit app	Live deployment for interactive project viewing.
Automated tests	24 tests passing in the project validation workflow.

10. Key Findings

For segmentation and retention analysis, the project treats customer-level outputs separately from aggregate sales metrics. This is important because customer intelligence features such as RFM scores, repeat-purchase patterns, cohort-style retention, and clustering require valid customer identifiers.

- The final validated completed-sales pipeline produced 1,007,913 completed-sales lines, GBP 20,476,260.45 in completed sales, 40,077 completed orders, and 11,205,148 units sold.
- The project identified 5,878 customers and 4,255 repeat customers, giving a repeat customer rate of 72.39%.
- RFM segmentation identified 1,290 Champions, 1,282 Hibernating customers, and 846 At Risk customers, giving clear groups for loyalty and reactivation actions.
- K-Means clustering showed strong revenue concentration: High-Value Champions represent 20.48% of customers but contribute 74.00% of revenue.

- The operational K-Means solution uses K=4 with silhouette 0.366, Davies-Bouldin 0.929, and five seed mean ARI 0.997, indicating stable customer clusters for business targeting.
- Customer-level analysis is limited by 243,007 anonymous rows (22.77% of raw lines), so aggregate sales analysis and customer-level segmentation are intentionally separated.
- The raw workbook contains 1,067,371 transaction lines, but 34,335 exact duplicates were detected; this equals 3.22% of the raw data and must be handled before publishing KPIs.
- December 2011 is a partial month, so it should be labelled and not compared directly with complete months.

11. Business Recommendations

- Protect the High-Value Champions cluster because it represents 20.48% of customers but contributes 74.00% of revenue; this group should receive loyalty recognition, priority support, and relevant retention offers.
- Use RFM-based reactivation campaigns for 1,282 Hibernating customers and 846 At Risk customers, with low-cost email or offer tests before heavier marketing spend.
- Because the repeat customer rate is 72.39%, retention should remain a core strategy; repeat buyers should be encouraged through replenishment prompts, cross-sell recommendations, and loyalty journeys.
- Because anonymous rows represent 22.77% of the raw data, the business should improve customer ID capture at checkout before relying heavily on customer-level segmentation.
- Because final signed returns/cancellations equal 7.14% of completed-sales value, product and return dashboards should be monitored separately from revenue dashboards.
- Because 34,335 exact duplicate records were found, every reporting refresh should include automated duplicate checks and reconciliation tests.
- Because December 2011 is incomplete, dashboards should mark it as a partial period and avoid month-over-month conclusions using that month.
- Because the dataset covers 5,304 products, product-level analysis should be used to identify high demand products, return-heavy items, and merchandising priorities.
- Since the dataset spans 43 countries, country-level revenue analysis should be used to prioritise international market focus and identify underperforming regions for targeted campaigns.

12. Dashboard and Deployment

Asset	Purpose
GitHub repository	Version-controlled project source code, documentation, and reproducible workflow.
Streamlit live app	Interactive project viewing and portfolio demonstration.
Power BI dashboard outputs	Executive-level reporting and visual summaries.
Reports and documentation	Explains data quality, scope, methodology, and interpretation limits.

13. Limitations

- The dataset does not contain customer demographics such as age, gender, income, or city-level location.
- Profit cannot be calculated because product cost is unavailable.
- The project does not claim verified churn prediction because no confirmed churn label exists in the dataset.
- Marketing campaign impact cannot be measured because campaign exposure data is unavailable.

- Historical customer value should not be treated as guaranteed future customer lifetime value.

14. Future Scope

- Add automated scheduled dashboard refresh when a production data source is available.
- Build a formal churn model only if validated churn labels or business-defined inactivity outcomes are provided.
- Add product category mapping for deeper merchandising analysis.
- Integrate marketing campaign data to measure conversion and retention impact.
- Improve the dashboard with drill-through pages for customer segment and product return analysis.

15. Conclusion

The Retail Customer Intelligence and Retention Analytics project successfully addresses the original business problem of converting raw retail transactions into a reliable customer intelligence layer. The validated pipeline identified GBP 20,476,260.45 in completed sales across 5,878 identified customers, 40,077 completed orders, and a 72.39% repeat customer rate.

The project also converted customer behaviour into actionable segments. RFM analysis highlighted Champions, At Risk, Hibernating, and other customer groups, while K-Means clustering showed that High-Value Champions represent 20.48% of customers but contribute 74.00% of revenue. These results directly support retention, loyalty, reactivation, and dashboard-based business decision-making.

My main learning from this project is that strong data science work is not only about building charts or models; it also requires data-quality validation, careful assumptions, clear business interpretation, and reproducible documentation. As an internship project, it demonstrates practical skills in data cleaning, analytical thinking, SQL modelling, feature engineering, visualization, customer analytics, dashboarding, testing, documentation, and project deployment.

References

- UCI Machine Learning Repository: Online Retail II dataset.
- Dataset DOI: 10.24432/C5CG6D.
- Pandas documentation for data manipulation and analysis.
- NumPy documentation for numerical operations.
- SciPy documentation for statistical analysis.
- Scikit-learn documentation for clustering and preprocessing workflows.
- Streamlit documentation for interactive web application deployment.
- Microsoft Power BI documentation for dashboard development and reporting.
- Project repository: github.com/ramakrishnareddy2112/Retail-Customer-Intelligence-Analytics.
- Live app: retail-customer-intelligence-analytics.streamlit.app.